

- 1 Let X be a set, μ^* be an outer measure on X , $\{A_j\}_{j \in \mathbb{N}}$ be a sequence of μ^* measurable disjoint sets, and $E \subset X$. We want to show that $\mu^*(E \cap (\bigcup_{i=1}^{\infty} A_j)) = \sum_{i=1}^{\infty} \mu^*(E \cap A_j)$. Note that

$$\mu^*(E \cap (\bigcup_{i=1}^{\infty} A_j)) = \mu^*(\bigcup_{i=1}^{\infty} (E \cap A_j)) \leq \sum_{i=1}^{\infty} \mu^*(E \cap A_j)$$

by the subadditivity of μ^* . Therefore we must show the inequality in the opposite direction. If we consider that for arbitrary j

$$\begin{aligned} \mu^*(E \cap (\bigcup_{i=1}^{\infty} A_i)) &= \mu^*(E \cap (\bigcup_{i=1}^{\infty} A_i) \cap A_j) + \mu^*(E \cap (\bigcup_{i=1}^{\infty} A_i) \cap A_j^c) \\ &= \mu^*(E \cap A_j) + \mu^*(E \cap (\bigcup_{i=1, i \neq j}^{\infty} A_i)) \end{aligned}$$

since $(\bigcup_{i=1}^{\infty} A_i) \cap A_j = \bigcup_{i=1}^{\infty} (A_i \cap A_j) = A_j$ as if $i \neq j$ then $A_i \cap A_j = \emptyset$ by definition and $(\bigcup_{i=1}^{\infty} A_i) \cap A_j^c = \bigcup_{i=1}^{\infty} (A_i \cap A_j^c) = \bigcup_{i=1, i \neq j}^{\infty} A_i$ since for all $i \neq j$, $A_j^c \cap A_i = A_i$. Therefore we have by induction that $\mu^*(E \cap (\bigcup_{i=1}^{\infty} A_j)) = \sum_{i=1}^n \mu^*(E \cap A_i) + \mu^*(E \cap (\bigcup_{i=n+1}^{\infty} A_j))$. Therefore $\mu^*(E \cap (\bigcup_{i=1}^{\infty} A_j)) \geq \sum_{i=1}^n \mu^*(E \cap A_i)$. Since this holds for all n , then it holds for the limit. Therefore we have equality.

- 2 Let μ^* be an outer measure on X with $\mu^*(X) < \infty$, and let $\nu^*(E) = \sqrt{\mu^*(E)}$ be defined for all $E \subseteq X$. To show that ν^* is an outer measure, we must show the following:

- $\nu^*(\emptyset) = \sqrt{\mu^*(\emptyset)} = \sqrt{0} = 0$
- If $B_1 \subseteq B_2 \subseteq X$ then we know that $\mu^*(B_1) \leq \mu^*(B_2)$. Therefore $\sqrt{\mu^*(B_1)} \leq \sqrt{\mu^*(B_2)}$, which implies that $\nu^*(B_1) \leq \nu^*(B_2)$
- Let $\{B_j\}_{j \in \mathbb{N}} \subseteq 2^X$. Then consider that

$$\begin{aligned} \nu^*(\bigcup_{j=1}^{\infty} B_j) &= \sqrt{\mu^*(\bigcup_{j=1}^{\infty} B_j)} \\ &\leq \sqrt{\sum_{j=1}^{\infty} \mu^*(B_j)} \\ &= \sqrt{\sum_{j=1}^{\infty} \nu^*(B_j)^2} \\ &\leq \sum_{j=1}^{\infty} \nu^*(B_j) \end{aligned}$$

Note that since all the values are positive and infinity is an acceptable value that the inequalities hold.

Now to show that A is ν^* measurable iff $\nu^*(A^c) = 0$ or $\nu^*(A) = 0$.

- ($\Rightarrow$). Suppose A is ν^* measurable. Then $\nu^*(X) = \nu^*(A) + \nu^*(A^c)$. Therefore,

$$\begin{aligned}\sqrt{\mu^*(X)} &= \sqrt{\mu^*(A)} + \sqrt{\mu^*(A^c)} \\ \mu^*(X) &= \mu^*(A) + \mu^*(A^c) + 2\sqrt{\mu^*(A)\mu^*(A^c)} \\ &\geq \mu^*(X) + 2\sqrt{\mu^*(A)\mu^*(A^c)} \\ \mu^*(X) &\geq \mu^*(X) + 2\sqrt{\mu^*(A)\mu^*(A^c)} \\ 0 &\geq 2\sqrt{\mu^*(A)\mu^*(A^c)}\end{aligned}$$

Note that this final inequality implies that either $\mu^*(A) = 0$ or $\mu^*(A^c) = 0$. Since these inequalities are maintained under the square root then the first direction has been shown.

- ($\Leftarrow$) Suppose WLOG that $\nu^*(A^c) = 0$. Then for arbitrary $E \subseteq X$, $0 \leq \nu^*(E \cap A^c) \leq \nu^*(A^c) = 0$ which implies that $\nu^*(E \cap A^c) = 0$. Therefore $\nu^*(E) \leq \nu^*(A \cap E) + \nu^*(A^c \cap E) = \nu^*(A \cap E) \leq \mu^*(E)$ which implies that $\nu^*(E) = \nu^*(A \cap E) + \nu^*(A^c \cap E)$.

3 Let μ^* be the Lebesgue outer measure, and $A \subset [0, 1]$. We must show that $1 = \mu^*(A) + \mu^*(A^c)$ iff A is Lebesgue measurable.

- ($\Rightarrow$) Suppose A is μ^* -measurable. Then $1 = \mu^*([0, 1]) = \mu^*(A) + \mu^*(A^c)$.
- ($\Leftarrow$) Suppose not. Then $1 = \mu^*(A) + \mu^*(A^c)$ and A is not μ^* -measurable. Therefore there exists $E \subset [0, 1]$ such that $\mu^*(E) \neq \mu^*(E \cap A) + \mu^*(E \cap A^c)$. We know by subadditivity that $\mu^*(E) \leq \mu^*(E \cap A) + \mu^*(E \cap A^c)$, therefore our inequality is strict with $\mu^*(E) < \mu^*(E \cap A) + \mu^*(E \cap A^c)$. If we assume our set E is μ^* -measurable then we have that

$$\begin{aligned}1 &= \mu^*(A) + \mu^*(A^c) \\ &= \mu^*(A \cap E) + \mu^*(A \cap E^c) + \mu^*(A^c \cap E) + \mu^*(A^c \cap E^c) \\ &> \mu^*(E) + \mu^*(A \cap E^c) + \mu^*(A^c \cap E^c) \\ 1 - \mu^*(E) &> \mu^*(A \cap E^c) + \mu^*(A^c \cap E^c) \\ \mu^*(E^c) &> \mu^*(A \cap E^c) + \mu^*(A^c \cap E^c)\end{aligned}$$

Note that by subadditivity that $\mu^*(E^c) \leq \mu^*(A \cap E^c) + \mu^*(A^c \cap E^c)$

Therefore we have a contradiction. Now assume that E is not measurable. Let the sequence E_n be the sets such that $E \subseteq E_n$, $E_n = \bigcup_{i=1}^{\infty} (a_i, b_i]$, $a_i < b_i$, $\mu^*(E_n) \leq \mu^*(E) + \frac{1}{n}$. Note we can always choose a cover which overshoots by at most $\frac{1}{n}$, as $\mu^*(E)$ is an infimum. Therefore we can choose $N \in \mathbb{N}$ such that $\frac{1}{n} < \mu^*(E \cap A) + \mu^*(E \cap A^c) - \mu^*(E)$. Therefore $\mu^* \leq \mu^*(E_N) \leq \mu^*(E) + \frac{1}{N} < \mu^*(E \cap A) + \mu^*(E \cap A^c) \leq \mu^*(E_N \cap A) + \mu^*(E_N \cap A^c)$, thus we have found a measurable set, E_N such that $\mu(E_n) < \mu^*(E_N \cap A) + \mu^*(E_N \cap A^c)$, and as we have shown previously is a contradiction. Therefore A must be Lebesgue measurable.

- 5 Let $E \subset [0, 1]$ with the property that for some $r \in (0, 1)$, every open interval I in $[0, 1]$ $\mu^*(E \cap I) \leq r\mu^*(I)$. Note that if we take $I = (0, 1)$ we have that $\mu^*(E) = \mu^*(E \cap I) \leq r\mu^*((0, 1)) = r$. Therefore $\mu^*(E) \leq r$. Since $\mu^*(E)$ is the infimum of the measures of the covers of E with open sets then there exists an open cover $\mu^*(\bigcup_{i=1}^{\infty} (a_i, b_i)) = r + \epsilon$ for some small $\epsilon > 0$. Therefore $\mu^*(E) = \mu^*(E \cap \bigcup_{i=1}^{\infty} (a_i, b_i)) \leq \sum_{i=1}^{\infty} \mu^*(E \cap (a_i, b_i)) \leq r(r + \epsilon)$. Note that $\epsilon \rightarrow 0$, therefore $\mu^*(E) \leq r^2$. If this holds up to n , where $\mu^*(E) \leq r^n$, then there exists an open cover where $E \subseteq \bigcup_{i=1}^{\infty} (a_i, b_i)$ and $\mu^*(\bigcup_{i=1}^{\infty} (a_i, b_i)) = r^n + \epsilon$. Therefore

$$\mu^*(E) = \mu^*(E \cap \bigcup_{i=1}^{\infty} (a_i, b_i)) \leq \sum_{i=1}^{\infty} \mu^*(E \cap (a_i, b_i)) \leq \sum_{i=1}^{\infty} r\mu^*((a_i, b_i)) = r^{n+1} + \epsilon r^n.$$

Since the choice of ϵ was arbitrary then $\mu^*(E) \leq r^{n+1}$. Since $0 < r < 1$, then $\lim r^n = 0$, thus $\mu^*(E) = 0$.